

Total No. of Questions : 8]

SEAT No. :

PE-4223

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[6583]-35

T.E. (Computer Engineering)

DATA SCIENCE AND BIG DATA ANALYTICS

(2019 Pattern) (Semester - VI) (310251)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q. 7 or Q.8.
- 2) Neat diagrams must be drawn whenever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary.
- 5) Use of Scientific Calculator is permitted.

Q1) a) What is the data Preparation phase in Data Analytics Lifecycle. What is the Analytics Sandbox and ETLT process in this phase? [8]

b) Compare the data acquisition phase with the data preprocessing phase in terms of inputs, outputs, and objectives. [8]

OR

Q2) a) List out the activities to be carried out in model planning and model building phase. What are different tools used for these phases? [8]

b) For a dataset of online retail transactions, which steps of the Big Data Analytics Lifecycle would you follow to build a recommendation system? [8]

Q3) a) What is logistic regression, and how does it differ from linear regression? What is the sigmoid Function, and what role does it play in logistic regression? [9]

b) Differentiate between predictive and descriptive analytics. What are NumPy and Pandas used for in Python? List any three essential Python libraries for data analysis [9]

OR

P.T.O.

- Q4) a)** Apply Naïve Bayes classification to a simple weather dataset to predict “Play/Tennis”. [9]

Day	Outlook	Play Tennis
1	Sunny	No
2	Sunny	No
3	Overcast	Yes
4	Rain	Yes
5	Rain	Yes
6	Rain	No
7	Overcast	Yes

- b) Explain how a decision tree splits data at each node with suitable example. Describe the role of entropy and information gain in determining the best split. [9]

- Q5) a)** Suppose you have the following dataset containing the coordinates of points in a 2-dimensional space : [9]

Point	X Coordinate	Y Coordinate
A	2	3
B	4	7
C	3	5
D	6	9
E	8	6
F	7	8

Perform K-means clustering on this dataset With $K = 2$. Assume the initial centroids to be (2,3) and (8,6). Compute the new centroids after each iteration until convergence, and assign points to their nearest centroids.

- b) How do you handle noise and irrelevant information in text data during preprocessing? Explain the terms bag of words and TF IDF in text analytics. [9]

OR

Q6) a) Explain the difference between agglomerative and divisive clustering. Compare single complete, and average linkage methods. [9]

b) What is the purpose of each of ROC, AUC and an elbow plot? Describe how each of these plots are constructed with suitable example. [9]

Q7) a) What is Bar chart, Pie chart and Line chart? Compare the use of bar charts and pie charts for categorical data. Analyze how visualization helps in identifying outliers. [9]

b) Compare HDFS and traditional file systems. Define NameNode and DataNode. Analyze how NameNode failure affects the Hadoop cluster. [9]

OR

Q8) a) What is a box plot? Explain the different components of a box plot? How do you interpret the median, quartiles, and whiskers in a box plot? What does the Interquartile Range (IQR) represent in a box plot? [9]

b) Explain why Pig is called a data flow language. Describe the role of the Pig execution engine. Explain why Hive is suitable for data warehousing. [9]

